Tasca S2.01. Nocions bàsiques SQL

Nivell 1

<u>Exercici 1</u>

- A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.

El esquema *transactions* está formado por dos tablas: *transaction* y *company*.

La tabla *company* contiene la información detallada sobre las empresas mientras que la tabla *transaction* contiene toda la información sobre las transacciones

La relación entre las tablas es de 1:N

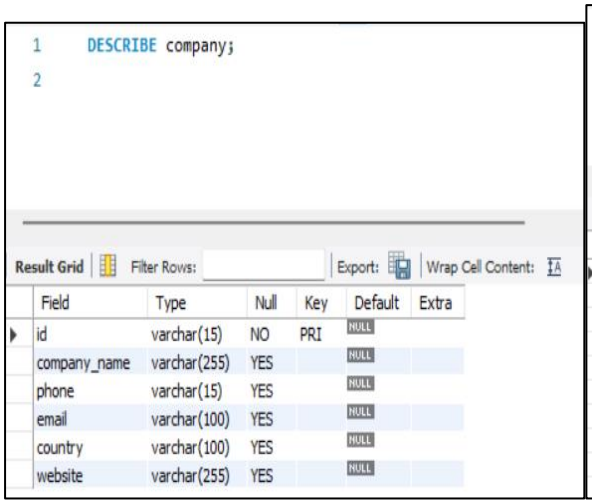
Cardinalidad: *company (1) ----- transaction (N)*

Tabla *company*

- Id (PK): Identificador único de compañía

Tabla *transaction*

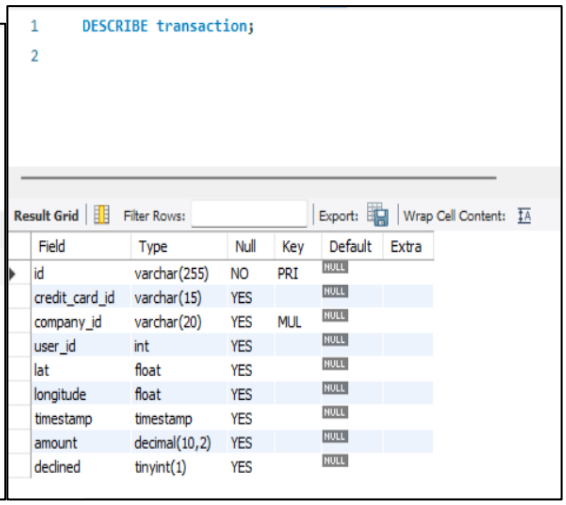
- Id (PK): identificador único de transacciones
- Company_id (FK): clave foránea que se relaciona con company.id



```
1    DESCRIBE company;
2
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	varchar(15)	NO	PRI	NULL	
company_name	varchar(255)	YES		NULL	
phone	varchar(15)	YES		NULL	
email	varchar(100)	YES		NULL	
country	varchar(100)	YES		NULL	
website	varchar(255)	YES		NULL	

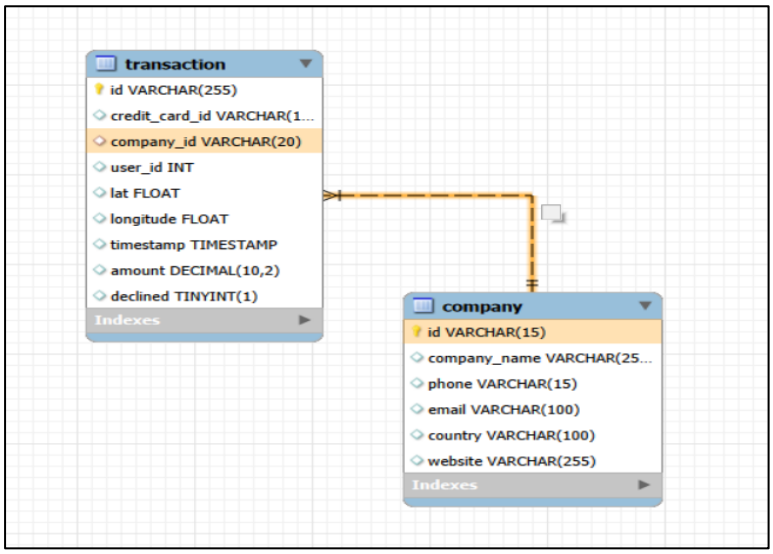


```
1    DESCRIBE transaction;
2
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	varchar(255)	NO	PRI	NULL	
credit_card_id	varchar(15)	YES		NULL	
company_id	varchar(20)	YES	MUL	NULL	
user_id	int	YES		NULL	
lat	float	YES		NULL	
longitude	float	YES		NULL	
timestamp	timestamp	YES		NULL	
amount	decimal(10,2)	YES		NULL	
declined	tinyint(1)	YES		NULL	

Diagrama

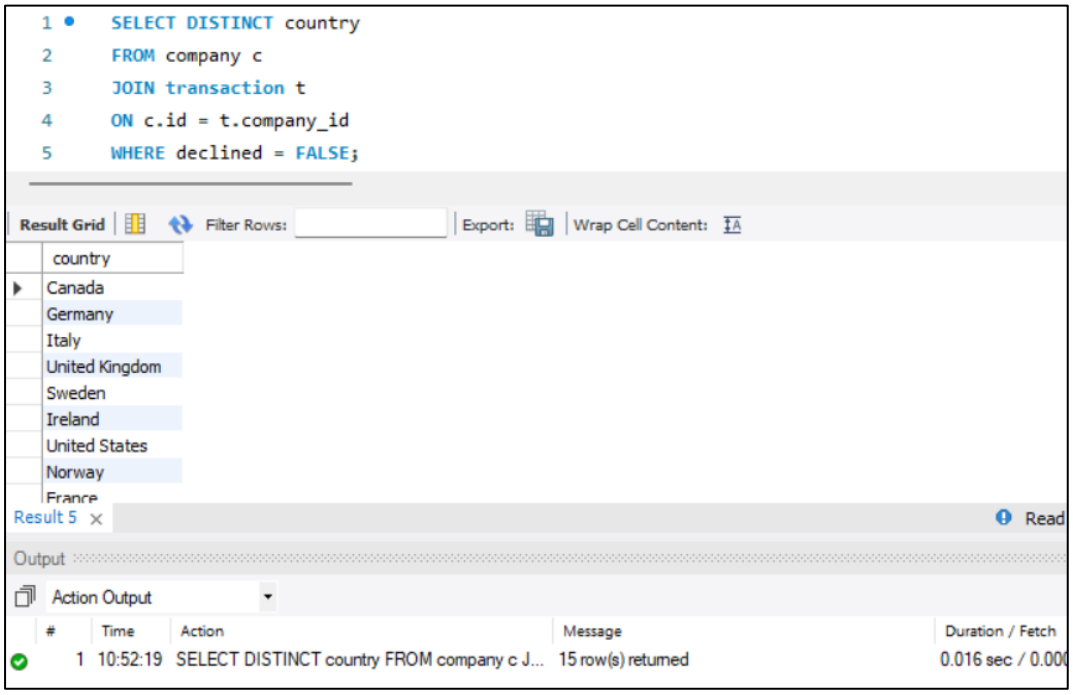


Exercici 2

Utilitzant JOIN realitzaràs les següents consultes:

- Llistat dels països que estan generant vendes.

Utilizo la cláusula WHERE declined = FALSE para tener en cuenta solo aquellas transcacciones que no han sido rechazadas.



```
SELECT DISTINCT country
FROM company c
JOIN transaction t
ON c.id = t.company_id
WHERE declined = FALSE;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country
Canada
Germany
Italy
United Kingdom
Sweden
Ireland
United States
Norway
France

Result 5 | Read

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:52:19	SELECT DISTINCT country FROM company c J...	15 row(s) returned	0.016 sec / 0.00

- Des de quants països es generen les vendes.

```sql
SELECT count(DISTINCT country) AS total_paises_con_ventas
FROM company c
JOIN transaction t
ON c.id = t.company_id
WHERE declined = FALSE;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_paises_con_ventas
15

Result 6

Output

Action Output

#	Time	Action	Message	Duration / F
1	10:55:08	SELECT count(DISTINCT country) AS total_pais...	1 row(s) returned	0.000 sec /

- Identifica la companyia amb la mitjana més gran de vendes.

```sql
SELECT company_name, round(AVG(amount),2) AS Media_ventas
FROM company c
JOIN transaction t
ON c.id = t.company_id
WHERE t.declined = FALSE
GROUP BY company_name
ORDER BY Media_ventas LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	Media_ventas
Augue Foundation	15.05

Result 7

Output

Action Output

#	Time	Action	Message	Duration /
1	10:56:41	SELECT company_name, round(AVG(amount),2)...	1 row(s) returned	0.000 sec

Exercici 3

Utilitzant només subconsultes (sense utilitzar JOIN):

- Mostra totes les transaccions realitzades per empreses d'Alemanya.

```sql
SELECT *
FROM transaction
WHERE company_id IN (
    SELECT id
    FROM company
    WHERE country = 'Germany');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
108B1D1D-5B23-A76C-55EF-C568E49A05DD	CcU-2938	b-2222	275	83.7839	-178.86	2021-07-07 17:43:16	293.57	0
EA2C3281-C9C1-A387-44F8-729FB4B51C76	CcU-2938	b-2222	275	20.2004	-116.84	2021-05-09 10:25:08	119.36	1
0DD2E608-5C9E-D1B3-4999-B99F43AD735A	CcU-2959	b-2234	275	9.68811	130.282	2021-04-17 05:30:17	252.47	1
AB069F53-965E-A2A8-CE06-CA8C4FD92501	CcU-2959	b-2234	275	1.64819	-158.007	2021-04-15 13:37:18	60.99	0
0466A42E-47CF-8D24-FD01-C0B689713128	CcU-4219	b-2302	170	-43.9695	-117.525	2021-07-26 07:29:18	49.53	0

transaction 9 | Apply

Output

Action Output

#	Time	Action	Message	Duration /
1	10:59:50	SELECT * FROM transaction WHERE company_id IN (SELECT id FR...	118 row(s) returned	0.000 sec

- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```sql
SELECT company_name
FROM company
WHERE id IN (
    SELECT company_id
    FROM transaction
    WHERE amount > (
        SELECT AVG(amount) AS Media_transacciones
        FROM transaction)
        );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name
Ac Fermentum Incorporated
Magna A Neque Industries
Fusce Corp.
Ante Iaculis Nec Foundation
Donec Ltd
Sed Nunc Ltd

company 14 | Read Only

Output

Action Output

#	Time	Action	Message	Duration /
1	11:05:02	SELECT company_name FROM company WHERE id IN (SELECT comp...	70 row(s) returned	0.000 sec

- Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```
SELECT *
FROM company
WHERE id NOT IN (
    SELECT company_id
    FROM transaction);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	company_name	phone	email	country	website
NULL	NULL	NULL	NULL	NULL	NULL

company 16 × Apply

Output

Action Output

#	Time	Action	Message	Duration /
1	11:09:02	SELECT * FROM company WHERE id NOT IN (SELECT company_id ...	0 row(s) returned	0.000 sec

Nivell 2

Exercici 1

- Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

Utilizo de nuevo el WHERE declined = FALSE para tener en cuenta solo las ventas no rechazadas

```
SELECT date(timestamp) AS fecha, sum(amount) AS total_ingreso
FROM transaction
WHERE declined = FALSE
GROUP BY fecha
ORDER BY total_ingreso DESC LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

fecha	total_ingreso
2021-12-20	1532.36
2021-04-22	1397.96
2021-05-09	1344.37
2022-02-26	1337.62
2021-03-29	1325.12

Result 17 ×

Output

Action Output

#	Time	Action	Message	Duration / F
1	11:13:25	SELECT date(timestamp) AS fecha, sum(amount) AS total_ingreso FROM ...	5 row(s) returned	0.000 sec /

Exercici 2

- Quina és la mitjana de vendes per país? Presenta els resultats ordenats de major a menor mitjà.

```
SELECT country, round(avg(amount),2) AS media_ventas
FROM company c
JOIN transaction t
ON c.id = t.company_id
WHERE declined = FALSE
GROUP BY country
ORDER BY media_ventas DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	media_ventas
United States	287.53
Ireland	285.83
Sweden	276.67
United Kingdom	271.77
Canada	261.94
Belgium	255.22

Result 19

Output

Action Output

#	Time	Action	Message	Duration / F
1	11:15:14	SELECT country, round(avg(amount),2) AS media_ventas FROM compa...	15 row(s) returned	0.000 sec /

Exercici 3

En la teva empresa, es planteja un nou projecte per a llançar algunes campanyes publicitàries per a fer competència a la companyia "Non Institute". Per a això, et demanen la llista de totes les transaccions realitzades per empreses que estan situades en el mateix país que aquesta companyia.

- Mostra el llistat aplicant JOIN i subconsultes.

```
SELECT *
FROM transaction t
JOIN company c
ON c.id = t.company_id
WHERE country = (
    SELECT country
    FROM company
    WHERE company_name = "Non Institute"
    );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined	id	company_
2B928E1C-EC14-A760-0A75-871477649D6A	CcU-2980	b-2246	275	-41.0496	161.685	2021-08-10 08:14:49	383.73	0	b-2246	Sed Nunc
ACD2011A-A2B1-C365-41E1-2AB00C65147A	CcU-2980	b-2246	275	-54.4792	-82.7974	2022-03-05 20:41:20	60.07	1	b-2246	Sed Nunc
4334349E-CEB0-3D68-A4D4-FEB7718A1ACE	CcU-3092	b-2310	275	-20.4859	150.87	2021-05-03 22:37:23	458.74	0	b-2310	Non Magna
BC2B9A38-77B4-28CD-1FE8-14DED863E773	CcU-3092	b-2310	275	-78.0295	18.5295	2021-10-18 07:27:35	477.95	1	b-2310	Non Magna
1479B3D2-B7BA-C7BB-4CE3-8D7C2DE85ABB	CcU-2994	b-2326	133	66.2672	172.399	2021-08-09 00:58:07	309.45	0	b-2326	Enim Cond

Result 20 | Read

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:17:05	SELECT * FROM transaction t JOIN company c ON c.id = t.company_id ...	100 row(s) returned	0.000 sec / 0.000

- Mostra el llistat aplicant solament subconsultes.

```sql
SELECT *
FROM transaction
WHERE company_id IN (
    SELECT id
    FROM company
    WHERE country = (
        SELECT country
        FROM company c
        WHERE company_name = "Non Institute")
    );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
2B928E1C-EC14-A760-0A75-871477649D6A	CcU-2980	b-2246	275	-41.0496	161.685	2021-08-10 08:14:49	383.73	0
ACD2011A-A2B1-C365-41E1-2AB00C65147A	CcU-2980	b-2246	275	-54.4792	-82.7974	2022-03-05 20:41:20	60.07	1
4334349E-CEB0-3D68-A4D4-FEB7718A1ACE	CcU-3092	b-2310	275	-20.4859	150.87	2021-05-03 22:37:23	458.74	0
BC2B9A38-77B4-28CD-1FE8-14DED863E773	CcU-3092	b-2310	275	-78.0295	18.5295	2021-10-18 07:27:35	477.95	1
1479B3D2-B7BA-C7BB-4CE3-8D7C2DE85ABB	CcU-2994	b-2326	133	66.2672	172.399	2021-08-09 00:58:07	309.45	0
152598C2-029D-D684-4B66-91EDF393EBFF	CcU-2994	b-2326	126	-67.0189	-141.672	2021-07-05 03:10:00	395.43	0
1B636B58-A2E8-7C69-D9C9-C54535DAED3B	CcU-2994	b-2326	131	70.2543	-13.1336	2021-07-06 08:48:46	195.06	0

transaction 21 × Apply

Output

Action Output

#	Time	Action	Message	Duration /
1	11:20:33	SELECT * FROM transaction WHERE company_id IN (SELECT id F...	100 row(s) returned	0.016 sec

Nivell 3

Exercici 1

- Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

No hay transacciones en las fechas indicadas

```sql
SELECT company_name, phone, country, date(timestamp) AS fecha, amount
FROM company c
JOIN transaction t
ON c.id = t.company_id
WHERE amount  BETWEEN 350 AND 400
AND date(timestamp) IN ('2015-04-29','2018-07-20','2024-03-13')
ORDER BY amount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	phone	country	fecha	amount

Result 23 × Read Only

Output

Action Output

#	Time	Action	Message	Duration / F
1	11:30:58	SELECT company_name, phone, country, date(timestamp) AS fecha, am...	0 row(s) returned	0.000 sec /

Exercici 2

- Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen més de 400 transaccions o menys.

```sql
SELECT company_name, count(t.id) as total_transacciones,
    CASE
        WHEN count(t.id) > 400 THEN 'Más de 400'
        ELSE 'Menos de 400'
    END AS Clasificación_por_transacciones
FROM company c
LEFT JOIN transaction t
ON c.id = t.company_id
GROUP BY c.company_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	total_transacciones	Clasificación_por_transacciones
Ac Fermentum Incorporated	2	Menos de 400
Magna A Neque Industries	2	Menos de 400
Fusce Corp.	2	Menos de 400
Convallis In Incorporated	2	Menos de 400
Ante Iaculis Nec Foundation	2	Menos de 400
Donec Ltd	2	Menos de 400
Sed Nunc Ltd	2	Menos de 400

Result 26 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / F
1	11:32:53	SELECT company_name, count(t.id) as total_transacciones, CASE WHE...	100 row(s) returned	0.016 sec /